

SYNOPSIS



On 10 August 2009, the Greek registered tanker *Saetta* and the Marshall Islands' registered tanker *Conger* collided when completing a ship to ship (STS) transfer operation off Southwold, Suffolk. *Saetta*'s starboard lifeboat and davit were damaged. *Conger* was undamaged and there were no injuries and no pollution.

The collision occurred at very slow speed, and resulted from the failure of *Conger*'s main engine to start as the vessels separated. Both manoeuvred to try and avoid a collision, but they were very close when the engine failed and the action taken was not effective. The response to the engine failure, and poor communications, were also contributory factors.

The number of STS operations off Southwold had increased considerably in 2009 and this was the third collision between ships involved in transfers in the area within a 6-week period. A further two collisions have occurred since. Although STS operations worldwide are reported to have a good safety record and the accidents off Southwold have been relatively minor, their frequency is cause for concern.

The Oil Companies International Marine Forum (OCIMF) has initiated the development of operational standards for STS service providers and occupational standards for STS superintendents, which will be published by mid 2010. In co-operation with the International Chamber of Shipping, OCIMF also intends to revise the STS Transfer Guide, to include operations between gas and chemical tankers, and advice on risk assessment and manpower requirements. In view of this action and the action taken by Fender Care Marine (FCM) and the operators of *Saetta* and *Conger*, no recommendations are considered necessary.

SECTION 1 - FACTUAL INFORMATION

1.1 PARTICULARS OF VESSELS AND ACCIDENT

Saetta

Registered owner	:	Nautica Shipmanagement S.A.
Technical operator	:	Cardiff Marine Inc
Port of registry	:	Piraeus
Flag	:	Greece
Type	:	Oil carrier, double hull
Built	:	2008 Shanghai, China
IMO number		9384069
Classification society	:	American Bureau of Shipping
Construction	:	Steel, double hull
Length overall	:	243.8m
Gross tonnage	:	58418t
Engine power and type	:	18420kW at 105rpm. B & W
Service speed	:	15.5kts
Persons on board	:	24
Injuries/fatalities	:	None
Damage	:	Impact damage to starboard lifeboat and davit arms

Conger

Registered owner	:	Conger Shipping Inc
Technical operator	:	Prime Marine Management
Port of registry	:	Majuro
Flag	:	Marshall Islands
Type	:	Oil / chemical tanker, double hull
Built	:	1999 Korea
IMO number		9198771
Classification society	:	Lloyd's Register
Construction	:	Steel, double hull
Length overall	:	228.6m
Gross tonnage	:	44067t
Engine power and type	:	16386kW at 99.5rpm. Sulzer 5 cylinder slow speed diesel
Service speed	:	14.5kts
Persons on board	:	23
Injuries/fatalities	:	None
Damage	:	None

Accident

Time and date	:	1241 UTC 10 August 2009
Location of incident	:	52° 12.15'N 001° 51.8'E 9.5miles south east of Southwold

1.3 NARRATIVE

1.3.1 Events preceding the STS transfer

Fender Care Marine Limited (FCM) was appointed by British Petroleum (BP) Shipping as the STS service provider⁴ to co-ordinate and facilitate the STS transfer between *Saetta* and *Conger* off Southwold. The company provided both vessels with information and instructions by e-mail to aid the planning and the execution of the proposed transfer. It also requested that its Checklist 1-Petroleum be completed and returned. On 4 August 2009, completed checklists were received from both vessels and FCM notified the Maritime and Coastguard Agency (MCA) of its intention to conduct the transfer (**Annex A**). The MCA advised that it would inspect *Conger* prior to the transfer. This was completed on 6 August, shortly after the vessel anchored off Southwold.

On 7 August 2009, checklist number one from the STS Transfer Guide; '*pre fixture information*', was completed by both vessels. The following day, *Saetta* carried out an STS transfer with the tanker *New Challenge* under the advisory control of a superintendent provided by FCM. The transfer was completed the next morning, and the vessels had separated by 1300.

At 1430, *Conger* rendezvoused with the tug *Flying Spindrift* approximately 4.5 miles south south east of Lowestoft (**Figure 1**). The superintendent, who had been on board *New Challenge*, and the fenders required for the STS operation, were then transferred from the tug to *Conger*. Under the control of the superintendent, the fenders were secured on the vessel's port side. On completion, the superintendent met with the deck officers to discuss the impending transfer. *Conger* then headed to the south-east and rendezvoused with *Saetta*. Communications between the superintendent on board *Conger* and *Saetta* was by VHF radio, channel 73.

1.3.2 Securing and cargo operations

Both tankers completed checklists 2 and 3 of the transfer guide, '*before operations commence*' and '*before run-in and mooring*' at 1710. *Conger* then closed the starboard side of *Saetta*, which was maintaining a steady course and speed. By 1848, *Conger* had secured to the starboard side of *Saetta* using 6 headlines, two forward springs, two aft springs and four stern lines. As the vessels prepared for the cargo transfer, the superintendent ordered *Saetta* to manoeuvre toward the agreed STS transfer position and anchor 1 mile clear of other shipping. At 1954, *Saetta*'s port anchor was dropped in position 52° 11.8'N 001° 58.0'E (**Figure 1**), and brought up with 7 shackles on deck. Checklist 4 of the transfer guide, '*before cargo transfer*', was then completed on both vessels. Cargo transfer operations commenced at 2012 and were completed at 0948 the following morning. Following the transfer of the cargo, *Saetta* had a freeboard of 9m, and *Conger* had a freeboard of 14m.

⁴ A service provider is a shore-based company employed to facilitate an STS operation

1.3.3 Preparations for sailing and unmooring

Shortly after cargo operations had finished, the superintendent passed a note to *Saetta*'s master apologising for the poor communications during the transfer with *New Challenge*. The note also explained his intentions for slipping the lines between *Conger* and *Saetta*, which were later reaffirmed during a face to face conversation over the bridge wings.

At 1215, a cargo surveyor on board *Saetta* called the superintendent and confirmed that the cargo operation was finished. Two minutes later, the superintendent advised *Saetta* that *Conger* was preparing her engines and that the unmooring operation would commence when her engines were ready. At 1235, *Saetta* reported that her engine would be on standby in 1 minute, but was requested to wait 15 minutes. The superintendent also advised that he would let the master of *Saetta* know when to send his crew fore and aft to attend the moorings.

At 1243, *Conger*'s main engine was tested ahead from the MCR and dead slow astern using the bridge telegraph. The engine started as intended on both occasions. Shortly afterwards, the superintendent briefed *Conger*'s deck officers on the intended sequence of events for slipping. On completion, *Saetta* commenced heaving in her anchor. This was completed at 1306.

The superintendent's plan was to manoeuvre from the anchorage and head in a north westerly direction, putting the wind on the port beam to help the vessels separate once all lines had been slipped.

With the vessels now underway on a heading of 031°, the superintendent ordered *Saetta* to put her helm hard to port and her engine to dead slow ahead. He then explained to *Saetta*'s master over the radio that he intended to '*single up as before*' referring to the recent separation from *New Challenge*, and advising that the last lines to be slipped would be two headlines from *Conger* and two stern lines from *Saetta*.

The vessels slowly turned to port, with their headings passing through 012° at 1313 when speed through the water had increased to 2.3 knots (**Figure 2**). The superintendent called *Saetta* and suggested that the time for completing transfer guide checklist 5 '*before unmooring*' be recorded as 1310. This checklist included '*The method of unberthing and of letting go moorings has been agreed with the other ship*'.

At 1316, *Saetta*'s heading was passing through 341° when the superintendent requested that she steady on a course of 330°. The unmooring operation commenced 2 minutes later.

At 1319, *Saetta*'s engine was stopped and 6 minutes later her master reported that he had lost steerage with the ship's head at 310°. More exchanges between the superintendent and the master of *Saetta* followed concerning which lines to slacken and slip. At 1327, the superintendent ordered *Saetta* 'dead slow ahead' and 'steady'.

1.3.4 The collision

After the last lines had been slipped, the superintendent ordered *Conger* 'starboard 5°' and for the main engine to be put to 'dead slow ahead'. Seconds later the engine was ordered to 'stop' and the helm increased to 'starboard 10°'. *Conger's* bow started to swing slowly to starboard, causing her stern to close towards *Saetta*.

To check this movement, at 1336 the superintendent ordered the engine to 'dead slow ahead', and the helm to 'amidships', quickly followed by 'port 5°'. The officer of the watch (OOW) placed the telegraph to dead slow ahead, and seconds later saw and heard the main engine alarm. He informed the chief engineer in the machinery control room (MCR) by telephone that the main engine alarm was sounding and that the engine had failed to start. He then shouted to the master on the bridge wing to inform him of the situation. The ship's officers spoke Ukrainian during these exchanges.

The master quickly went inside the bridge and discussed the situation with the OOW. Meanwhile, the superintendent was aware that *Conger* was starting to fall astern of *Saetta*, and ordered 'port 10°' and the main engine to 'slow ahead'. The helmsman applied the helm as ordered, but when the OOW placed the telegraph to slow ahead, the engine again did not start. The master returned to the bridge wing and advised the superintendent that '*the engine not starting*'.

At the chief engineer's request, the OOW put the engine telegraph to 'stop' followed by 'dead slow ahead'. The main engine again failed to start. The telegraph was returned to 'stop' and the master ordered the OOW not to touch it. He also instructed the forward mooring team to leave the forecastle.

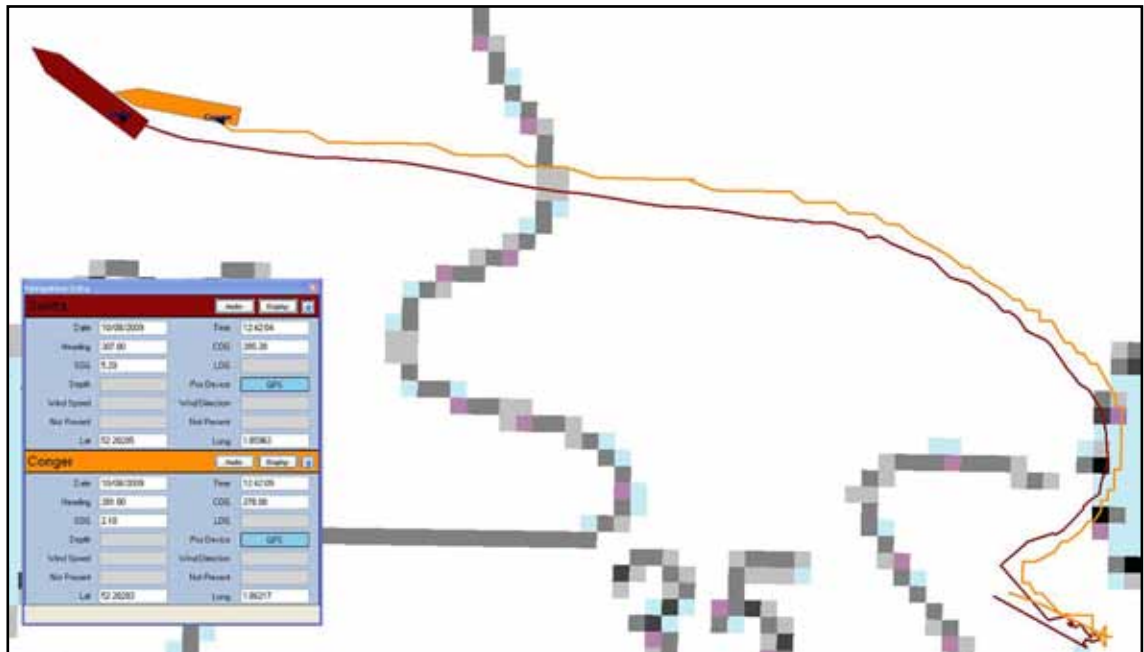
At 1337, the superintendent ordered *Saetta* 'dead slow ahead' and 'easy to starboard' to try and increase the lateral distance between the sterns of the vessels. In response, *Saetta's* master used varying amounts of helm up to 30°. The superintendent also ordered *Conger's* helm to 'amidships' to arrest the continuing slow swing to port. Seconds later, the superintendent broadcast on VHF radio '*lost our engines we have lost our engines*'. There were no ship names given before or after the message, and the bridge team on *Saetta* did not hear it. The superintendent then repeated his order for *Saetta* to proceed at 'dead slow ahead' and 'easy starboard'.

At 1338, the master asked the OOW what had happened to the engine. The OOW again telephoned the chief engineer in the MCR and was informed that the problem was being investigated. One minute later, the chief engineer called the bridge and asked for permission to test the engine. This was refused by the master.

Conger's bow continued to swing extremely slowly towards *Saetta* and, at 1340, the superintendent ordered 'dead slow astern'. The engine telegraph was moved as instructed and the engine started. The superintendent immediately

ordered 'emergency full astern', quickly followed by 'stop' 'hard to starboard' and 'full ahead'. The OOW missed the final order, but immediately advised the superintendent 'engine stopped'. The superintendent then ordered 'slow ahead' 'full ahead' 'hard to starboard' 'stop' and 'full astern' in less than 1 minute. Seconds later, at 1342, *Conger's* port anchor struck the starboard lifeboat of *Saetta*, and her port bulwark hit the lifeboat davit arm. At impact, *Conger's* heading was 289° and her speed was estimated to be less than 1 knot (**Figure 4**). The superintendent quickly moved inside *Conger's* bridge and ordered the main engine to 'stop' and the helm to 'amidships'.

Figure 4



AIS tracks of vessels (collision)

Just prior to the collision, the superintendent had assessed that *Saetta's* pivot point was forward of *Conger's* bow, and ordered *Saetta* to 'full ahead' and 'hard to starboard' in an attempt to manoeuvre her clear. *Saetta's* master remained unaware of *Conger's* engine problems and had started to issue his own helm and engine orders to try and avoid a collision, but he had not informed the superintendent of his actions.

1.3.5 Post accident actions

The vessels soon separated and anchored close to the position of the collision. The superintendent reported the collision to Yarmouth coastguard.

Saetta suffered structural damage to her starboard lifeboat (**Figure 5**) and the lifeboat davit arm (**Figure 6**). *Conger* suffered only scratches to her paintwork (**Figure 7**).



Scratches to *Conger's* paintwork

Conger received approval from the Marshall Islands administration and Lloyd's Register to proceed to her next port, and she departed UK territorial waters at about 1900 on the day of the accident. Following approval from the Greek administration to temporarily use additional liferafts in lieu of the damaged lifeboat, *Saetta* remained off Southwold in readiness for further STS operations.

1.4 ENVIRONMENTAL CONDITIONS

The wind was south westerly 6 to 9 knots. The sea state was smooth and visibility was good. The predicted tidal stream was setting south west at 1.1 knots and there was at least 30m of water under the vessels' keels.

1.5 MAIN ENGINE

Conger was fitted with a five cylinder slow speed reversing engine. Similar to many slow speed diesel engines, the engine had to be stopped before engaging ahead or astern, and was connected to a fixed pitch propeller.

When *Conger* arrived at Piraeus, Greece, 10 days after the accident, an electronic systems engineer inspected the engine control system. The inspection found that the system was functioning correctly.

International

International regulation for the 'Prevention of Pollution during Transfer of Oil Cargo between Oil Tankers At Sea' will come into force on 1 January 2011 via an amendment to the International Convention for the Prevention of Pollution from Ships (MARPOL) (**Annex C**). The amendment provides general rules on safety and environmental protection, including the requirement for any oil tanker involved in STS operations to carry on board an STS plan approved by the vessel's administration. The plan is to be developed using industry best practice, which the regulation identifies to be the OCIMF / ICS STS Transfer Guide.

The forthcoming regulation also places responsibility on vessels to notify coastal states when it is intended to carry out a ship to ship transfer within territorial waters, or EEZ at least 48 hours before the operation is scheduled to commence.

1.14 SIMILAR ACCIDENTS

1.14.1 Southwold

The MAIB is aware of four other collisions that occurred off Southwold between 3 July and 1 December 2009:

On 3 July, a bunkering vessel had completed its operation with a tanker, and was unmooring when her starboard quarter struck the port wing ballast tank of the tanker, causing an indentation.

On 25 July, two tankers similar in size to *Saetta* and *Conger* were preparing to come alongside for an STS transfer within UK territorial waters. As the vessels closed, the manoeuvring vessel had difficulty maintaining a parallel heading, and the approach was aborted. During a second approach, the helmsman of the manoeuvring vessel had difficulty in maintaining the ordered heading as the vessels closed; the manoeuvring vessel started to swing towards the constant heading vessel despite the use of full opposite rudder. The superintendent was not made aware of the steering difficulties, but took action to try and avoid a collision, including manoeuvring the constant heading vessel. The vessels collided, with the port bow of the manoeuvring vessel striking and penetrating number 3 starboard ballast tank of the constant heading vessel.

Contributory factors identified included: the master's lack of STS experience, the probable effects of interaction, poor communications on the bridge of the manoeuvring vessel, the lack of a risk assessment, the lack of a pilot card, poor line of sight between the bridge wing gyro repeater and bridge instrumentation, and an inability to secure a forward secondary fender at deck level.

On 27 August, an STS transfer involving a VLCC (constant heading vessel) and an Aframax tanker (manoeuvring vessel) resulted in a collision just outside UK territorial waters during the final stages of the approach. The superintendent ordered the constant heading vessel to reduce to dead slow ahead. Seconds later, the stern of the manoeuvring vessel started to close the stern of the VLCC. The superintendent ordered the helm hard to port and the main engine slow ahead. He was informed that the engine was operating dead slow astern. Before any action could be taken to recover the situation, the vessels collided. The hull of the VLCC suffered a 6m dent on the starboard side, and the hull of the manoeuvring vessel suffered 12 sq m of deck plate buckling, considerable damage to stiffeners below the deck, and damaged railings and davits.

Contributory factors included poor communications, and the inability to secure the aft secondary fender accurately in a position to protect the vessel in line with the guidance provided in the ICS / OCIMF guide. The superintendent in control of the transfer was also in control of the operation conducted on 25 July, and has been recommended for refresher training.

On 1 December 2009, a collision occurred when two vessels were in the final stages of their approach. The helmsman, who had been on the wheel for approximately 1.5 hours, was given an order to apply starboard helm to counter the manoeuvring vessel's swing to port. The vessel continued her swing, and it was only moments before impact that it became apparent that port helm had mistakenly been applied. Contributory factors to the accident were: the operation had been delayed and was conducted in the hours of darkness using a bridge wing rudder indicator that was not visible from the superintendent's conning position; the OOW was not monitoring the helmsman's actions; and the superintendent was too pre-occupied with other aspects of the operation to recognise that his orders had not been complied with.

1.14.2 Accident statistics

Accident data provided by FCM showed that of its 2688 STS operations carried out worldwide in 2008, there were 6 collisions and 5 near misses reported. In 2009, up until the end of August, of its 2260 STS operations carried out worldwide, 6 collisions and 6 near misses were reported. Through its membership of Intertanko, FCM, along with the oil majors, attempted to establish a global database of accidents during STS transfers. However, the database was not populated by other service providers, and was discontinued.

SECTION 2 - ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 ENGINE FAILURE

The sequence of events leading to the collision between *Conger* and *Saetta* was triggered by the failure of *Conger*'s main engine to start as the vessels separated. Subsequent investigation identified that an air start pilot valve on the engine's number three cylinder was not working correctly.

Following the initial ahead movement used to separate the bows after the vessels had unmoored, it is almost certain that the engine had been stopped in the position in which the number three cylinder was the first cylinder to fire when the engine was next ordered ahead. As a result, when dead slow ahead was ordered, air was unable to pass through the defective pilot valve and the engine did not start.

The engine started when dead slow astern was ordered because in this direction number three cylinder would not have been the first cylinder to fire, and the pilot valves of all other cylinders were functioning correctly. Engine failure had been identified as a hazard by FCM in its risk assessment of STS operations (**Table 1**), and had relied on the effective vetting of vessels as its control measure. However, as the pilot valves had been maintained in accordance with the manufacturer's instructions, the engine had been tested before unmooring was commenced, and there had been no previous indication of the fault, the defect was difficult to foresee, and had possibly been latent for some time.

However, the actions taken by the OOW, master and chief engineer following the failure did not follow a predefined procedure. This resulted in a 4.5 minute delay in restoring propulsion, which occurred by chance when the superintendent ordered 'slow astern'. It is likely that, had a breakdown procedure been available and followed, the main engine would have been started sooner.

2.3 COMMUNICATIONS

Conger's main engine failed to start when the superintendent was attempting to arrest the movement of the vessel's stern towards *Saetta*. This was a critical time, but prompt and effective action was hindered by two significant breakdowns in communications.

First, although the failure was quickly seen by *Conger*'s OOW, it was reported to the master in Ukrainian and the superintendent was not informed until about 1 minute later.

Second, *Saetta*'s master was never aware of the engine failure prior to the collision because: the superintendent did not follow basic voice procedures when broadcasting his message on VHF radio; *Saetta*'s bridge team did not hear the superintendent's transmission; and no emergency signal was sounded.

Co-ordinating the movements of two vessels underway in very close proximity requires clear and concise communications between all parties, but this accident and the accidents summarised in paragraph 1.14 indicate that this is not always achieved. The need for good communications is particularly relevant when the persons involved are working in a second language. As in this case, the possibility of messages being missed or misunderstood and persons forgetting protocols or reverting to their native language when under pressure is inevitably increased. It is therefore essential that during STS operations, account is taken of the language and communication channels to be used, and the roles of the persons involved, when determining the composition and organisation of bridge teams.

2.4 DECISION-MAKING

The departure manoeuvre had gone as planned, until the engine failure, but by the time the superintendent had been informed of the problem, *Conger* had started to turn very slowly back towards *Saetta*, which was increasing speed and moving ahead. The superintendent was very experienced in STS operations, but with the vessels probably no more than 30m apart, he was placed in a very difficult situation, which he was unable to control.

The very slow turn to port had been initiated by the use of the rudder and was possibly assisted by the wind acting on the vessel's port side accommodation, and by the forces of interaction acting between the vessels. However, although the helm had been ordered to 'amidships' shortly before the engine failure, no counter helm was applied to check the movement towards *Saetta* until the engine had been started.

Once the engine was started, the superintendent issued a series of orders in rapid succession, one of which was missed by the OOW and others which were not given sufficient time to have an effect. This resulted in the vessel continuing to make slow headway. Given *Conger*'s slow speed at impact, it is highly likely that, had *Conger*'s engine been kept running continuously astern since it had restarted, the collision might have been avoided. Although *Saetta*'s master acted independently to try and avoid collision, this action was taken too late to have any effect.

This was the superintendent's eighth and final STS transfer within a 20 day period. Notwithstanding two periods of rest ashore, this had been an arduous period of employment, with rest periods dictated by operational requirements and prevailing weather conditions. This had led to disrupted sleep patterns over

the course of his contract and a lack of sleep during the 48 hours before the collision. However, although the superintendent reported that he did not feel tired, it was inevitable that he was suffering from the effects of fatigue; it is not possible to determine to what extent, if any, this affected his behaviour under pressure, his judgment, or his manoeuvring of the vessels.

2.5 PLANNING AND BRIEFING

In preparation for unmooring, the superintendent briefed *Conger's* master and OOW in person. His briefing to *Saetta's* master was by a note indicating that it would be conducted in a similar way to the STS transfer with *New Challenge*, followed up by a face to face conversation over the bridge wings. No information was passed regarding the times to have main engines ready, or weigh anchor, and there were no details provided regarding the intended unmooring and departure manoeuvres. Therefore, given *Saetta's* master's inexperience in command during STS operations, it was not surprising that the superintendent had to repeat his instructions, or that the master had to request clarification.

The risks involved when two vessels unmoor and separate while underway are potentially the same as when coming together and mooring. Therefore, it is important that both mooring and unmooring operations be planned in detail, and that all aspects of these operations be briefed. However, this does not appear to have been identified in the risk assessments undertaken by the service provider or the vessels. In this case, the superintendent appears to have directed the unmooring operation and departure manoeuvre as events unfolded rather than following a specific plan. In addition, the absence of a comprehensive briefing to both masters indicates that the superintendent was largely content to rely on the completion of checklist 5 – '*before unmooring*' to signify that both vessels were ready. The absence of a comprehensive brief was also an opportunity missed to clarify the actions to be taken in the event of an engine or steering failure on either vessel, including the signal to be sounded on the vessel's whistle.

Although the checklists in the ICS / OCIMF guide are a useful 'aide mémoire' to ships' crews, robust operational risk assessments are also required to determine the depth and detail of the actions required. Moreover, when checklists are used frequently, a 'tick-box' mentality can develop, which seriously undermines their effectiveness. This possibly explains why neither master was aware of the emergency signal to be used, and the master of *Saetta* signed the unmooring checklist as completed, even though he had not been fully briefed.

2.6 THE SERVICE PROVIDER

FCM's operations manual was comprehensive and was largely in accordance with guidance in the STS transfer guide. The company had also been proactive in designing ship-handling simulator training for its superintendents. However, a number of shortcomings identified indicate that its safety management system would benefit from some revision.

In particular, the operations manual placed the onus on its superintendents to ensure they were sufficiently rested; the service provider had not taken any steps to monitor the hours of work and rest of its superintendents as advised in the ICS / OCIMF Guide. The potential for the superintendent to become fatigued would have been considerably decreased had the superintendent's hours of work and rest also been monitored by the service provider in accordance with the guidance in the ICS / OCIMF Guide, and designated rest periods planned into his work schedule.

In addition, the superintendent's performance had not been assessed annually as suggested in the STS transfer guide. Such assessments help to monitor competence and compliance with operational procedures. Importantly, they also help to identify if additional training is required.

FCM had been vetted by the oil majors it provided with STS services. In the absence of any global operational standards for service providers, each oil major vetted the company against its own criteria, and did not share the results of the vetting. Inevitably, this required FCM to meet differing requirements demanded by each oil major. It is almost certain that the provision of operational standards for service providers within the STS guide would not only help to simplify the vetting process for the majors, but it would also provide service providers with a benchmark standard to reach and maintain.

2.7 ROLE OF THE SUPERINTENDENT

The growth of STS operations worldwide, and the diversity of the products transferred, have resulted in the participation of many crews who are not experienced in this activity. In turn, this has led to the reliance on superintendents to assist inexperienced masters. However, although the employment of superintendents in these circumstances is advised in the ICS / OCIMF guide, this does not absolve any master from his responsibility for the safety of his vessel. In this respect, the decision of *Saetta's* master to take action to try and avoid a collision was appropriate.

It is evident from **Annex B** that the responsibilities of a superintendent in STS operations are wide-ranging, time-consuming, and demanding. With regard to manoeuvring, his role is similar to that of a harbour pilot, particularly when embarked on a tanker approaching or separating from a vessel at anchor. It is even more demanding when neither vessel is at anchor.

When such operations are conducted by two vessels underway, a superintendent is expected to provide direction and advice to the masters of both vessels. Potentially this can be a challenging task which can be influenced by: the experience and training of the superintendent; the opportunities to rest; the experience of the vessels' masters in STS operations; the ability and performance of the bridge teams; and potential language or communications difficulties. In this case it was made far more difficult when one vessel had an emergency which seriously affected her manoeuvrability.

SECTION 3 - CONCLUSIONS

3.1 SAFETY ISSUES IDENTIFIED DURING THE INVESTIGATION WHICH HAVE BEEN ADDRESSED

1. When *Conger's* main engine failed to start when ordered ahead, there was no procedure in place for the crew to follow. This possibly led to the engine remaining stopped for about 4.5 minutes. [2.2]
2. Prompt and effective action following the loss of *Conger's* main engine was hindered by two significant breakdowns in communications. It is essential during STS operations, that account is taken of the language and communication channels to be used and the roles of the persons involved when determining the composition and organisation of bridge teams. [2.3]
3. The unmooring operation and departure manoeuvre was not properly planned or briefed, and there was an over-reliance on checklists. [2.5]
4. The provision of operational standards for service providers would help to simplify the vetting process for the oil majors and would provide service providers with a benchmark. [2.6]
5. The service provider had not taken any steps to monitor the hours of work and rest of its superintendents, or assessed the superintendent's performance annually as suggested in the STS transfer guide. [2.6]
6. The responsibilities of a superintendent in STS operations are wide-ranging, time-consuming, and demanding, and vary according to the type of transfer conducted. The risk of overload could be reduced by the provision of a second superintendent or an assistant. [2.7]
7. There are no occupational standards defined for superintendents or masters in control of STS operations. [2.8]
8. It is important that the ICS / OCIMF STS Transfer Guide (Petroleum) is reviewed and amended to fully reflect current operations worldwide and to take into account the lessons learned from this and other recent accidents. [2.9]